



Vibration Control of the NASA Curiosity Rover Landing Mechanism
Control Systems

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INTRODUCTION

The Curiosity rover is a car-sized robotic rover exploring Gale Crater on Mars as part of NASA's Mars Science Laboratory mission (MSL). Curiosity transformed from its stowed flight configuration to a landing configuration while the MSL spacecraft simultaneously lowered it beneath the spacecraft descent stage with a 20 m (66 ft) tether from the “sky crane” system to a soft landing—wheels down—on the surface of Mars. The goal is to design a control system that will limit the oscillations of the rover, in order to allow for high-quality images from the landing camera that is used to navigate the MSL “sky crane”. The designed control system includes a vibration system as shown in Fig. 1. The vibration system includes a mass-spring-damper system. Moreover, it includes a linear actuator that is acting on the mass of the vibration control system. The vibration control system is placed inside the Curiosity rover. The simplified system of the “sky crane” and the rover including the vibration system are shown in Fig. 2.

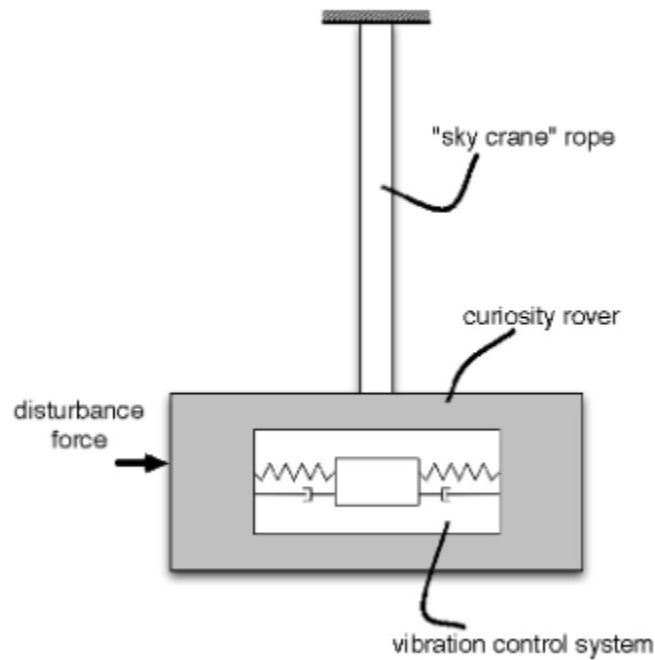


Figure 1: Simplified model of the “sky crane” system and the Curiosity rover.

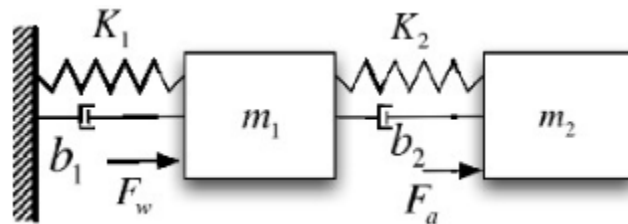


Figure 2: Simplified model of the “sky crane”, the rover and the vibration control system

SYSTEM PARAMETERS

Table 1: The system parameters are shown below where m_1 is the the mass of the rope and the rover, b_1 is the damping of the rope, K_1 is the stiffness of the rope, and m_2 , b_2 , and K_2 is the mass, damping, and stiffness of the vibration control system, respectively. Assume $L = 7$, $F = 5$, and $A = 32$.

$$m_1 = 5L$$

$$m_2 = 0.7L + 0.1A$$

$$K_1 = 1000A$$

$$b_2 = 8F + 2L$$

$$b_1 = 0.9F$$

$$K_2 = 10F + 50L$$

EQUATIONS OF MOTION

Equations 1a and 2a were formulated using the Impedance method in the Laplace domain. F_w and F_A represent the disturbance force and the actuator force, respectively.

$$X_1(m_1 s^2 + (b_1 + b_2)s + K_1 + K_2) - X_2(b_2 s + K_2) = F_w \quad [1a]$$

$$-X_1(b_2 s + K_2) + X_2(m_1 s^2 + b_2 s + K_2) = F_w \quad [2a]$$

After plugging in the system parameters, taking the inverse Laplace, and isolating the second derivative terms, the following equations of motion, Equation 1 and Equation 2, are found.

$$\ddot{x}_1 = \frac{1}{35}F_w - \frac{117}{35}\dot{x}_1 - \frac{6480}{7}x_1 + \frac{54}{35}\dot{x}_2 + \frac{80}{7}x_2 \quad [1]$$

$$\ddot{x}_2 = \frac{10}{81}F_A + \frac{20}{3}\dot{x}_1 + \frac{4000}{81}x_1 - \frac{20}{3}\dot{x}_2 - \frac{4000}{81}x_2 \quad [2]$$

STATE-SPACE REPRESENTATION

The state-space representation for the equations of motion are shown below.

State Variables

$$X = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \end{bmatrix} \quad \dot{X} = \begin{bmatrix} \dot{x}_1 \\ \ddot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_2 \end{bmatrix}$$

State Equation

$\dot{X} = AX + BU$, $Y = CX + DU$ where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-6400}{7} & \frac{-117}{35} & \frac{80}{7} & \frac{54}{35} \\ 0 & 0 & 1 & 0 \\ \frac{4000}{81} & \frac{20}{3} & \frac{-4000}{81} & \frac{-20}{3} \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ \frac{1}{35} & 0 \\ 0 & 0 \\ 0 & \frac{10}{81} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad U = \begin{bmatrix} F_W \\ F_A \end{bmatrix}$$

And $D = 0$

MATLAB SIMULATION

The A, B, C, and D matrices were inputted into MATLAB to simulate the motion uncompensated system. All MATLAB code is put into Appendix A. Figure 3 below displays the system response caused by a unit step disturbance.

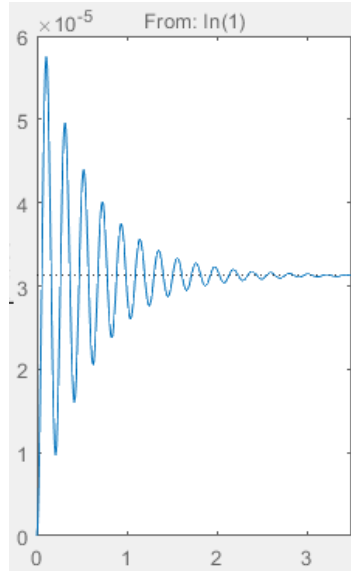


Figure 3: Plot for unit step disturbance for uncompensated system generated by MATLAB

DESIGNING A CONTROLLER

Simulink was used to model the system using a block diagram, seen in Figure 4 below. A PD controller was used to optimize the settling time. The gains for the P and D components were changed until the minimum settling time was achieved. This was done by increasing P and D in small increments until the settling time increased. Most values of P and D would cause the system to settle so slowly that Simulink would not display a settling time for it.

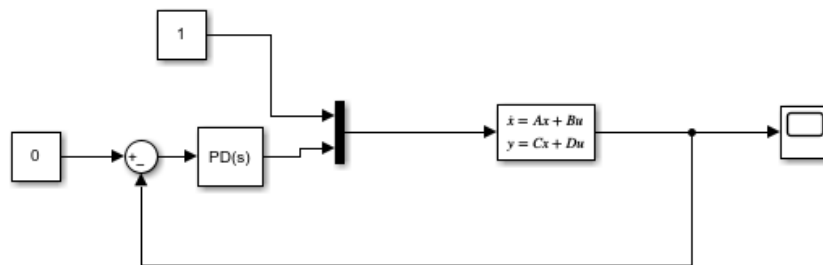


Figure 4: Simulink Block Diagram of the compensated system

The final gains were $K_p = 1$ and $K_d = 0.01$ for the proportional and derivative controller, respectively. The setting time was 2.255 seconds. The plot for the system response is shown in Appendix B.

OPTIMIZATION OF VIBRATION CONTROL PARAMETERS

If the controller fails, the system can still be optimized by changing the parameters of the vibration control system: the stiffness, damping, and mass. This was done on MatLab because the system can be represented with the open-loop transfer function. m_2 , b_2 , and K_2 were modified until the `stepinfo()` command displayed a small settling time. The final vibration control parameters were $m_2 = 15$, $b_2 = 100$, and $K_2 = 320$ with a rise time of 1.3162 seconds. The plot of the system is shown in Figure 5 below.

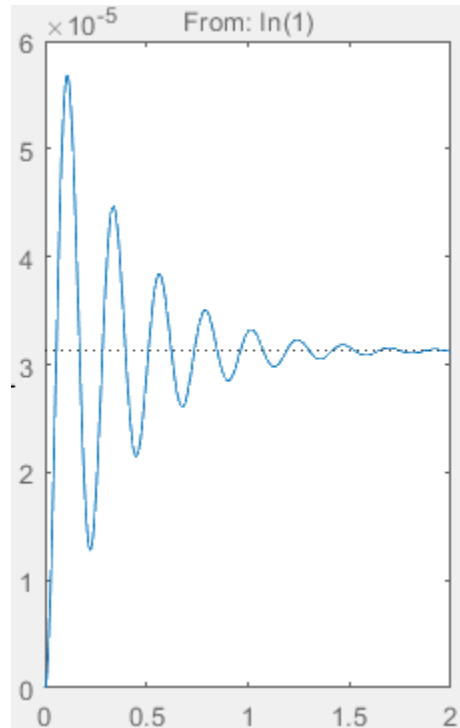


Figure 5: Plot for unit step disturbance with optimized control system parameters generated by MATLAB

APPENDIX A: MATLAB CODE AND OUTPUTS

Figure 1A below shows the code that simulates the uncompensated system using a state space representation in MATLAB.

```

1 %The state space is represented by the matrices below.
2 - A = [0 1 0 0; -6480/7 -117/35 80/7 54/35; 0 0 0 1; 4000/81 20/3 -4000/81 -20/3];
3 - B = [0 0; 1/35 0; 0 0; 0 10/81];
4 - C = [1 0 0 0]
5 - D = [0 0];
6 - set_param('ControlsProj/State-Space','A','A','B','B','C','C','D','D')
7
8 % Create the state space system
9 - sys = ss(A,B,C,D);
10
11 % Simulate the system
12 - step(sys)
13

```

Figure 1A: MATLAB code of the state space representation of the uncompensated system

Figure 2A below shows the code that simulated the system that optimizes the vibration control parameters. It also includes the Command Window output of the code.

```

1 %The system parameters are represented below
2 - F = 5 %Letters in first name
3 - L = 7 %Letters in last name
4 - A = 32 %Sum of digits in UDID
5 - m1 = 5*L %Mass of the sky crane rope and the rover
6 - k1 = 1000*A %Stiffness of the sky crane rope
7 - b1 = 0.9*F %Damping of the sky crane rope
8 - m2 = 15 %Mass of the Vibration System
9 - k2 = 100 %Stiffness of the Vibration System
10 - b2 = 320 %Damping of the Vibration System
11
12 %The state space is represented by the matrices below.
13 - A = [0 1 0 0; -(k1+k2)/m1 -(b1+b2)/m1 k2/m1 b2/m1; 0 0 0 1; k2/m2 b2/m2 -k2/m2 -b2/m2];
14 - B = [0 0; 1/m1 0; 0 0; 0 1/m2];
15 - C = [1 0 0 0]
16 - D = [0 0];
17 - set_param('ControlsProjPartB/State-Space','A','A','B','B','C','C','D','D')
18
19 % Create the state space system
20 - sys = ss(A,B,C,D);
21
22 % Simulate the system
23 - step(sys)
24 - ST=stepinfo(sys)
25 - ST(2)
26

```

```

RiseTime: 0.0552
SettlingTime: 1.3162
SettlingMin: 2.0945e-05
SettlingMax: 4.5885e-05
Overshoot: 46.8309
Undershoot: 0
Peak: 4.5885e-05
PeakTime: 0.1455

```

Figure 2A: (top) MATLAB code of the state space representation of the optimized system.
(bottom) Output of the MATLAB Code

APPENDIX B: SIMULINK INPUTS AND OUTPUTS

Figure 1B below shows the inputs for the PD Controller and State Space blocks in Simulink. The MATLAB code was used to load matrices A, B, C, and D, into the state space block.

The image shows two Simulink parameter input forms. The left form is for a PD Controller, with fields for 'Controller parameters', 'Source' (set to 'internal'), 'Proportional (P):' (set to '1'), and 'Derivative (D):' (set to '0.01'). The right form is for a State Space block, with fields for 'Parameters' (A, B, C, D) and 'Initial conditions:' (set to '0').

Controller parameters	
Source:	internal
Proportional (P):	1
Derivative (D):	0.01

Parameters	
A:	A
B:	B
C:	C
D:	D
Initial conditions:	0

Figure 1B: (left) Input for the PD controller block. (right) Input for the State Space block.

Figure 2B below shows the output of the scope in the block diagram. It shows the system response using a PD controller.

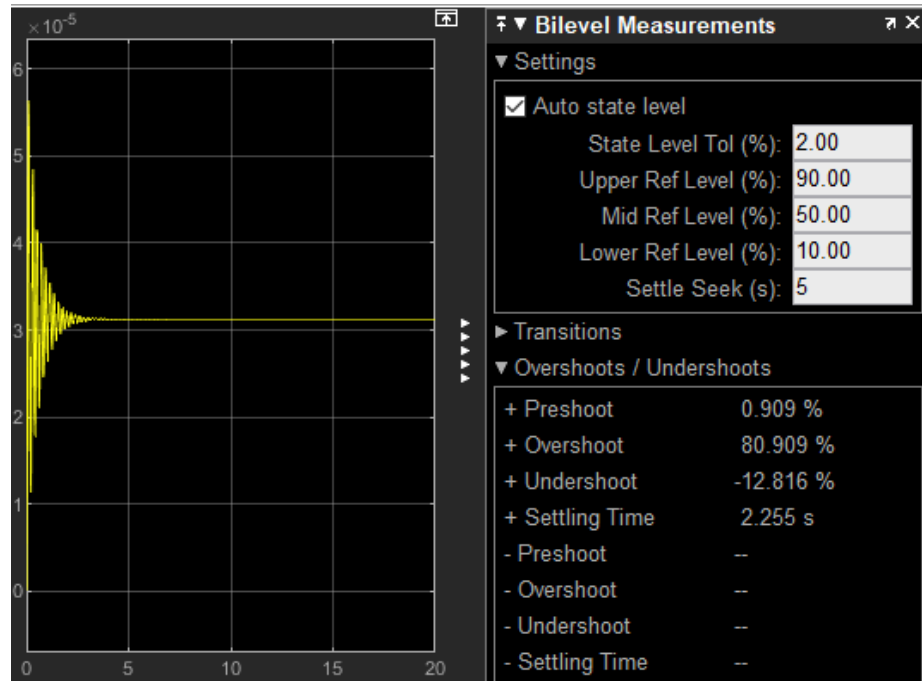


Figure 2B: Simulink plot and response information for the controlled system.